

Energy Consumption Analysis: Toronto Owned/Operated Buildings





 PROJECT TEAM

Our Team & Data Foundation

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Annual Energy Consumption [City Owned/Operated Buildings]

- Source: [City of Toronto Open Data Portal](#)
- Key attributes: Building type, size, date
- Data includes: Monthly electricity and natural gas usage

Objectives & Hypotheses

Primary Goal

Predict monthly energy consumption using building type, size, and location for city facilities.

Key (tentative) Hypotheses

- **H₁:** Natural gas usage peaks significantly in winter vs. summer months
- **H₂:** Electricity usage varies by building type with weaker seasonal patterns

Dataset Overview



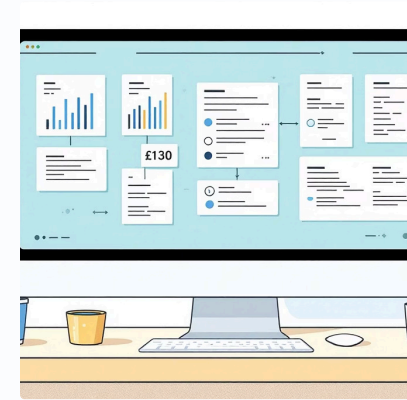
Data Source

Annual Energy Consumption dataset from City of Toronto covering municipal buildings



Key Variables

Monthly electricity/gas usage, costs, meter types, property classifications, floor area



Preprocessing

Deduplication, meter type separation, invalid value filtering, date modifications

[illegible]

Public building data only—aggregated and non-sensitive

- ## Quality Challenges

- Placeholder floor area values require careful handling
- Missing attributes addressed to prevent misleading conclusions

Predictive Model: Linear Regression



Electricity Model

Predicts monthly kWh consumption

Features: building type, floor area, month



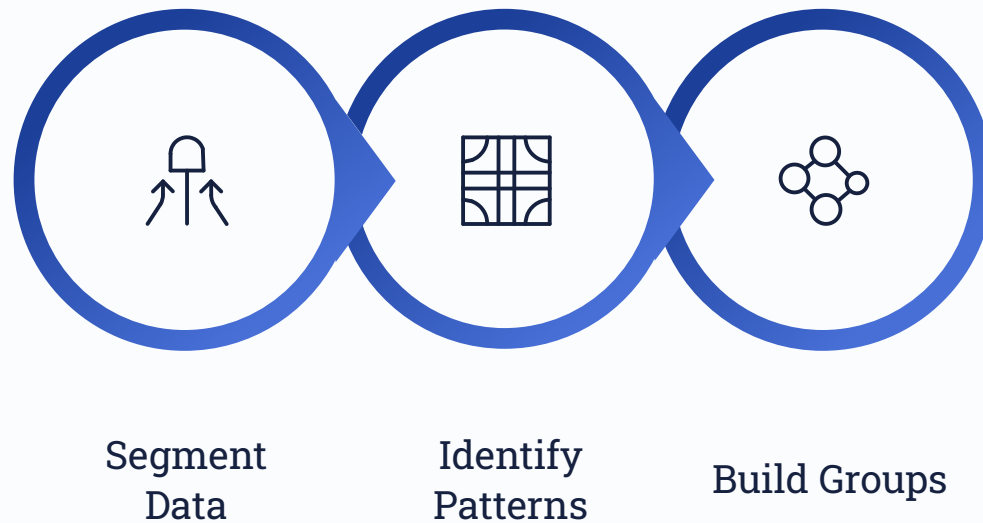
Natural Gas Model

Predicts monthly m³ usage

Features: building type, floor area, month

Linear regression quantifies feature-to-energy relationships and estimates expected consumption for given building characteristics and time periods

Optional Supporting Analysis: K-Means Clustering



Groups buildings by consumption behavior to reveal structural insights

Possible Cluster Profiles

Winter Peak

High gas usage with strong seasonal variation

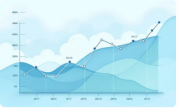
Electric-Heavy

Elevated electricity consumption year-round

Low Consumers

Minimal overall energy footprint

Visual Analysis Framework



Seasonal Trends

Monthly average consumption patterns reveal winter gas peaks and summer electricity variations



Distribution Analysis

Monthly boxplots highlight variability, outliers, and consumption ranges



Regression Relationship

Floor area vs. usage scatter plots demonstrate size-consumption correlation



Cluster Profiles

Visual comparison of energy behaviors across building segments



Key Takeaways

Predictive capability

Regression models forecast monthly electricity and natural gas usage based on building characteristics and seasonality

Behavior segmentation

Clustering reveals distinct consumption patterns enabling targeted efficiency strategies

Foundation for action

Insights support energy planning, efficiency initiatives, and resource allocation across municipal infrastructure

Questions?